Project Development — Model Performance Testing

Date	_____
Team ID	_____
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Model Performance Testing

The project team shall fill the following information in the model performance testing template. This documents key metrics, filter usage, visualization count, and any issues encountered during testing.

S.No.	Parameter	Screenshot / Values
1	Data Rendered	EV Dataset — Total Records: ~10,000 charging sessions across 15 models
2	Data Preprocessing	Cleaned columns: model, battery_capacity_kWh, range_km, temp_celsius, terrain, charge_time_min, session_date
3	Utilization of Filters	Filters applied: Vehicle Model, Battery Size, Temperature Range, Terrain Type, Date Range
4	Calculation Fields Used	Real Range vs. Advertised Range, Charge Efficiency %, Range Drop by Temperature, Battery Degradation Index
5	Dashboard Design	No. of Visualizations: 7 (Range Heatmap, Charge Curve, Model Comparison Bar, Temp Scatter, Degradation Line, Summary Stats, Export View)
6	Performance Results	All charts loaded within 1.8 seconds. Filters responded within 0.5 seconds. No crashes on tested browsers (Chrome, Firefox, Edge).
7	Test Devices	Desktop (Windows 10, Chrome 120), Laptop (macOS, Safari 17), Mobile (Android, Chrome Mobile)
8	Issues Found	Minor: Temperature filter reset did not update heatmap on first click — fixed in Sprint-3 patch.

Summary of Visualizations

- Range Heatmap — Geographic view of range depletion per trip
- Charge Curve — Battery % vs. charging time per session
- Model Comparison — Side-by-side bar chart of real range by model
- Temperature vs. Range — Scatter plot showing cold/heat impact
- Degradation Trend — Line chart of battery capacity over charge cycles
- Summary Stats Panel — Total trips, avg range, avg charge time
- Export View — Clean chart view for PDF/PNG download